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TWEAKER HELP OVERVIEW

organized by Dorado

HELP CENTER UPDATE VERSION: 0918.2019

NEWS

The Latest Product Update

NEW

Tweaker Metal ECO /
Timing ECO Cell Stealing

NEW

Tweaker HPC /
Dominant Scenario Analysis

NEW

Tweaker Attribute (Procedure) System /
Timing ECO QoR Summary :
QoR_collector.tcl

UPDATE

Tweaker Clock ECO /
Clock Surgery

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GUI Operation Related

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Design Related
(Netlist / Schematic View Related)

02

Violation Related
(Slack / Path View Related)

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Timing Related
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Physical Related
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FAQ

TWEAKER HELP NEWS

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Tweaker Metal ECO / Timing ECO Cell Stealing

License Category: Tweaker-M1

08.31.2019

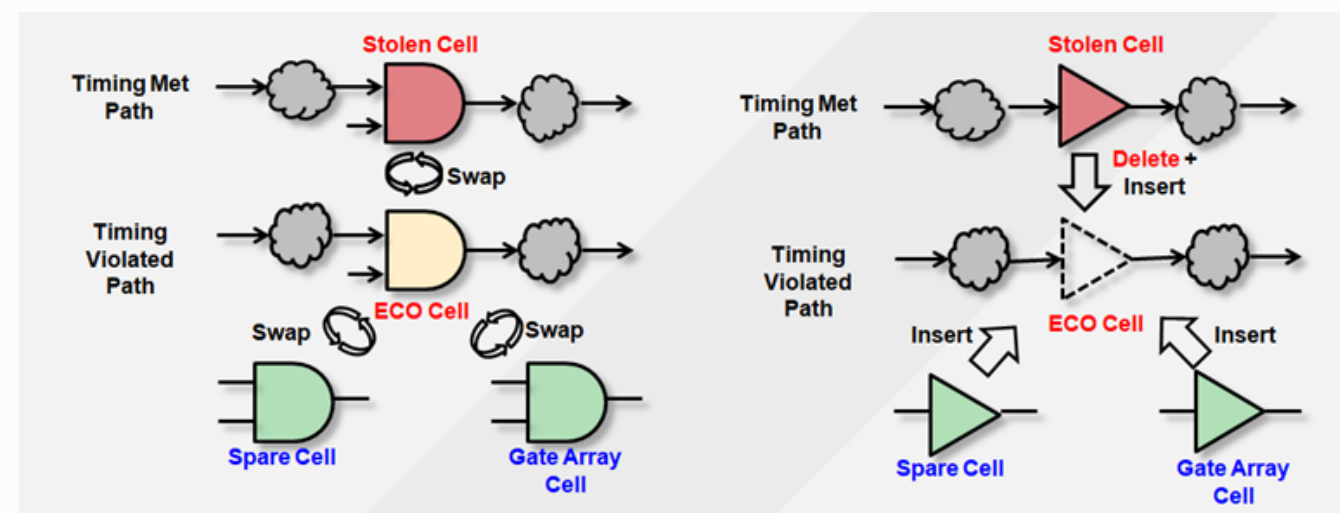
Cell Stealing For Metal Timing Closure

During post-mask Tapeout, users usually only have SpareCells, GateArrayCells, and ECOFiller can be reused. If there is no enough SpareCell's resource, Metal ECO will be very difficult to converge. Therefore, Tweaker provides another solution "StealingCells" to reduce users' manual effort and increase the success rate in Metal ECO.

StealingCells is checking the existing function cells which have enough timing margin and then switching to SpareCells for timing-critical paths reusing. At the same time, the same function of SpareCells which can't use for timing-critical paths, they will be reused for non-timing critical paths. This is the concept about switching existing Cells and SpareCells for timing driving purpose, and this will increase the resource of metal ECO.

Tweaker has been working on Metal ECO solution for more than a decade. Not only SpareCells, GateArrayCells, and ECOFillerCells, but Tweaker also supported ContactCells and RecycleCells. In this Tweaker version, tool provides auto SteallingCells solution. More resource will make postMaskECO easier, and more automation will make the metal eco process faster.

For more information, please check with your local Tweaker support.





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Tweaker HPC / Dominant Scenario Analysis

License Category: HPO

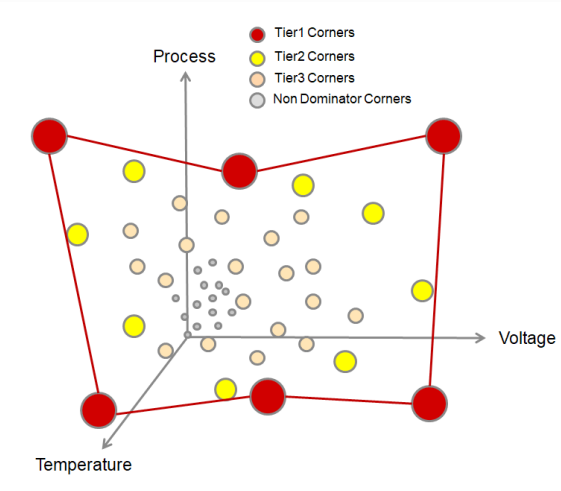
04.09.2019

Find The Dominate Scenarios from Hundred Sign-off Corners

Under the situation of processing numerous scenarios at the tape-out stage, running all corners in ECO tools is time-consuming. To solve this issue, conventional methods handpick critical timing corners based on STA reports, which is a tedious and time-demanding job. Even worse, it does not guarantee to cover the majority of critical violations. To address the difficulty, Tweaker not only automatically analyzes sign-off STA data but also utilizes library corner scaling technique to extract the most critical ones.

By reading MMMC timing violation report and timing window files, Tweaker will do the RC/Library Scaling to find out the most critical corners base on setup and hold slack report. It then categorizes scenarios into different priority levels as below.

- Tier 1 corners: scenarios that must be read
- Tier 2 corners: scenarios that are recommended to monitor
- Tier 3 corners: scenarios that can be ignored



	Setup Check	Hold Check
Original Corners	24	54
Tier 1 Corners	5	4
Tier 2 Corners	4	6
Tier 3 Corners	15	44

Base on DSA result, your ECO run can easily reduce number of scenarios, and only focus on timing critical corners to speed up runtime and reduce ECO iteration.

For more information, please check with your local Tweaker support.



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Tweaker Attribute (Procedure) System / Timing ECO QoR Summary : QoR_collector.tcl

License Category: Tweaker-T1

08.31.2019

Timing ECO QoR Summary

The user always needs to know what' s the summary of ECO information before and after Tweaker ECO. Through Tweaker attribute commands and plus the procedure of TCL commands, we provide customize of QoR Summary command which can provide design information, timing ECO information, power ECO information, and ECO output information.

After Tweaker reads all inputs data and creates ECO domain (by the command "slkdc -check_slack_consistency"), user can use Tweaker command "QoR_collector -input tweaker.log -output ./outputDir/ " to check all ECO related information. The first is design information, it has the number of library information(lib/lef), the number of design information (including netlist/def), and the number of scenario information (including spef/sdf/twf). The 2nd is violation information, including before and after ECO how many FEP, TNS, WNS, and total violation paths change. The 3rd is power information, including leakage, internal, dynamic power data. The 4th is ECO output information, including where the eco script and Tweaker db be generated.

Those design QoR summary data can be generated in csv or html format, so user can attach and sent it to user directly to review how the ECO QoR in their end.

For more information, please check Tweaker procedure as below or your local Tweaker support.

\$TWEAKER_INSTALL/dorado/etc/template/twk_utilities/standalone/QoR_collector.tcl



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Tweaker Clock ECO / Clock Surgery

License Category: Tweaker-C1

08.31.2019

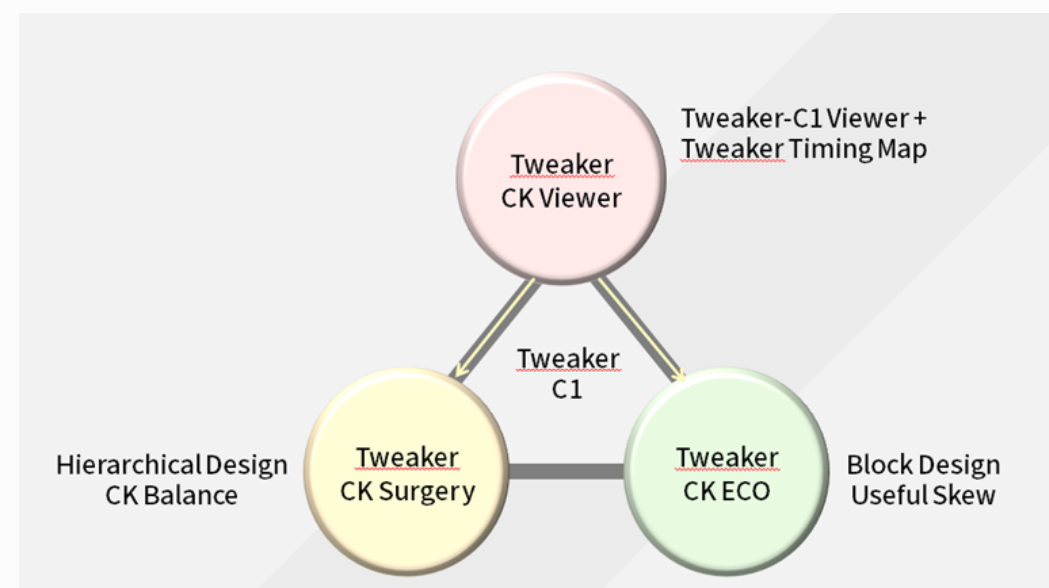
Clock Surgery for Clock ECO

Clock Tree Synthesis (CTS) is one of a milestone for chip implementation. Good CTS can reduce not only ECO iteration but also deliver good design quality. However, traditional CTS stage in P&R tool has some limitations, like the gape of STA correlation and only a few numbers of MMMC support. Therefore, how to help the user to do incremental CTS optimization is one of the important tasks for us.

Since Tweaker already has the structure to consider sign-off STA timing and do partial changes in the design, the tool can read full clock expanded report which is provided by more sign-off corners in the latest P&R database. Based on STA sign-off report, Tweaker will be able to modified clock paths and update timing graph to all the other related paths. Just like regular data path ECO, clock surgery can quick update result if there is any timing impact.

For Tweaker clock surgery, there are two solutions user can do. One is clock skew to borrow the timing from different level and minimize the worst setup violations in ECO stage. The other is clock balance which read hierarchical design and helps the user to easily catch and adjust trees to minimize the number of violation count in the early stage. User can do all types of manual ECO base on STA sign-off report to surgery clock tree and easy to achieve their tapeout goal.

New feature clock surgery can help the user to manually analysis tree and adjust the tree quickly. Within this small clock changes, it can speed up timing converge and avoid redundant cell usage and power consumption.





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Design Related (Netlist / Schematic View Related)

What would you like to know?

- ▶ How to Trace Logic Connectivity in Schematic View
- ▶ How to Do Function ECO in Schematic View



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Violation Related (Slack / Path View Related)

What would you like to know?

- ▶ How to Open Timing Violation Path
- ▶ How to Review Blocking Code for Timing Violation
- ▶ How to Check Blocking Code in Detail
- ▶ How to Relax Blocking Code for Timing Fixing



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Timing Related (Slack / Path View Related)

What would you like to know?

- ▶ How to Check Timing Margin
- ▶ How to Check the Worst Scenario
- ▶ How to Check Timing Violation in Specified Slack Range



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Physical Related (Physical View Related)

What would you like to know?

- ▶ How to Highlight Timing Path in Physical View
- ▶ How to Zoom-in and Zoom-out in Physical View
- ▶ How to Highlight Available Resource
- ▶ How to Highlight Detail Route in Physical View
- ▶ How to Highlight Detour Net Physical View
- ▶ How to highlight Physical Rule Violation



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Power Related (Physical View Related)

What would you like to know?



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Frequently Asked Questions

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Log Message

What would you like to know?

- ▶ How to Check ECO Environment Setting
- ▶ How to Check ECO QoR in Each Iteration
- ▶ How to Check Detail about Tweaker Blocking Code or Error Code



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Frequently Asked Questions

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Script Information

What would you like to know?

- ▶ How to Check Detail in Tweaker Command and Variable
- ▶ How to Read Timing Violations Which Only Related to BlockB
- ▶ How to Apply Don't Touch Setting
- ▶ How to Fix Interface Timing Violation

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For other ECO question or Tweaker usage, please contact with tweaker_support@dorado-da.com or visit our [E-Learning webpage](#)



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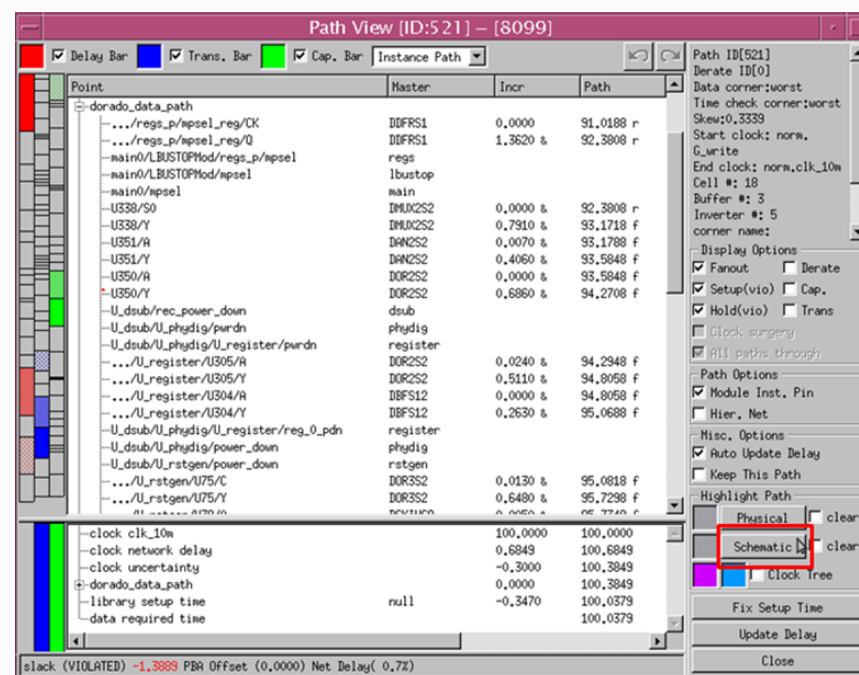
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Design Related / How to Trace Logic Connectivity in Schematic View

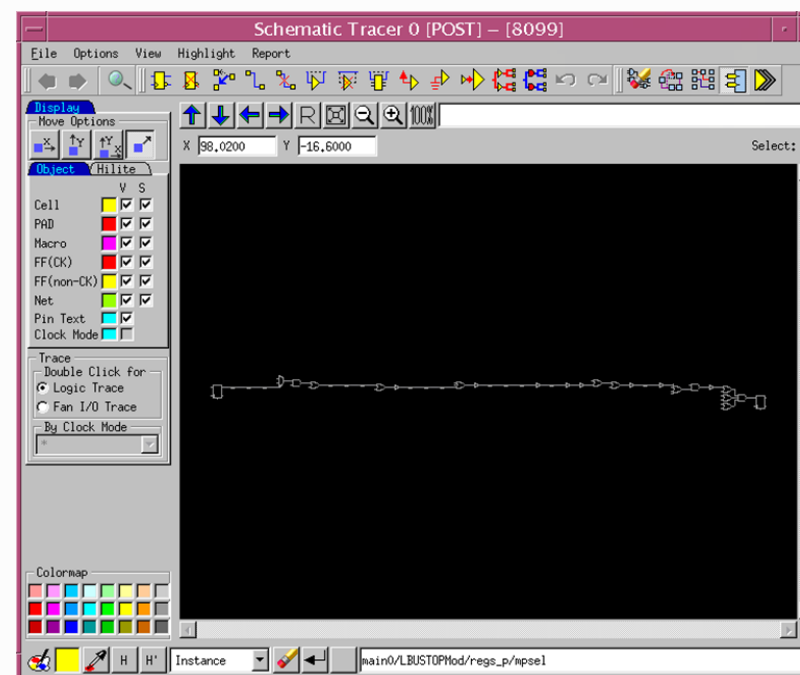
License Category: Tweaker-T1

09.02.2019

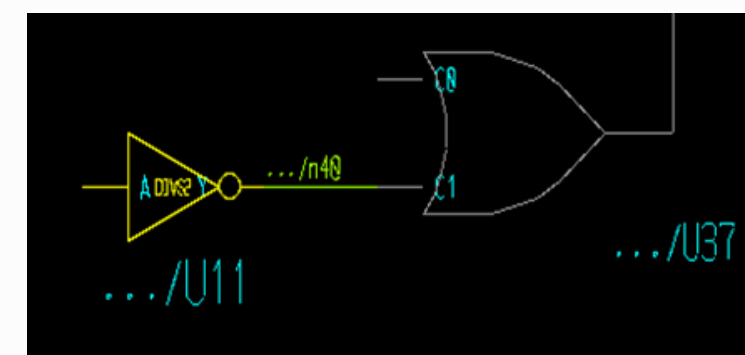
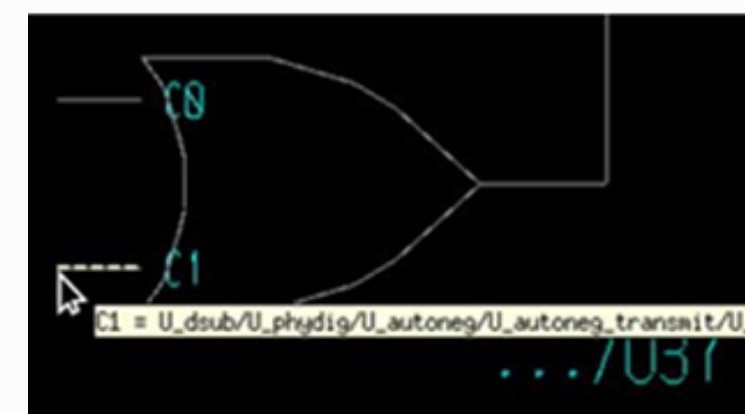
① When opening “Path View”, the user can check the logic connection for this path by opening “Schematic View”.



② Path Logic Connection in “Schematic View”.



③ User can check the connectivity for each cell by double left click on the instance's input or output pin.





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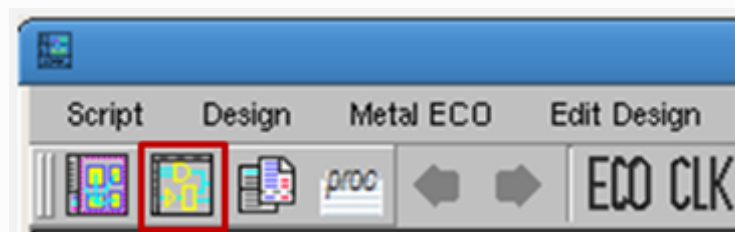
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Design Related / How to Do Function ECO in Schematic View

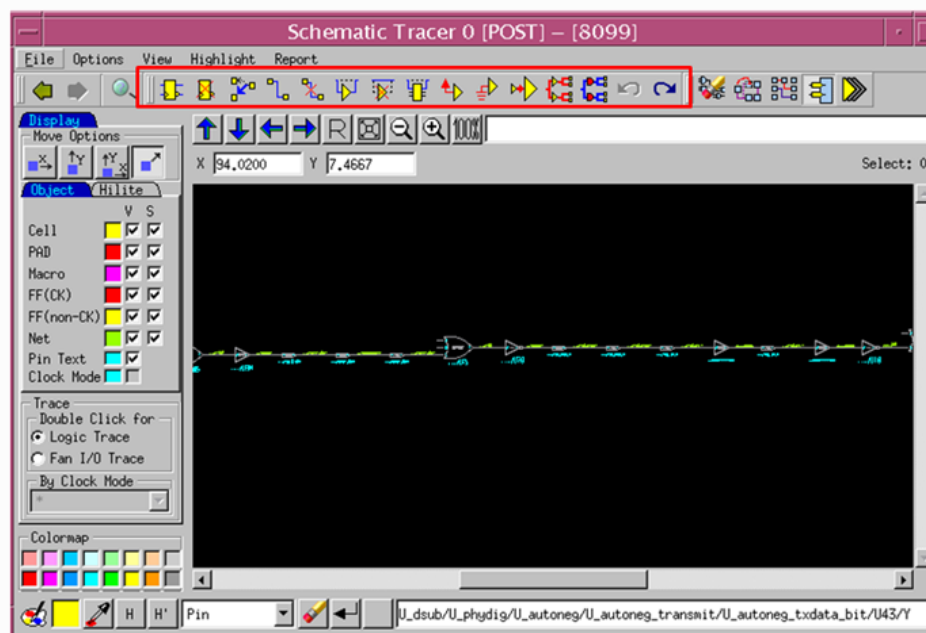
License Category: Tweaker-T1

09.02.2019

- ① Press “Schematic View” icon in netlist(main) view to start functional ECO.

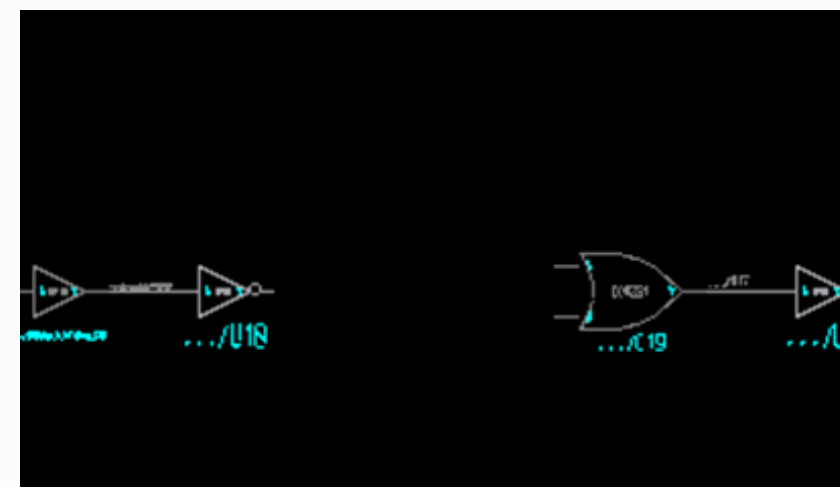


- ② Drag, press, and trace the logical ECO cone.



- ③ Delete Instance :

Click  and select instance from “Schematic View” .
Instance would be removed after selected.



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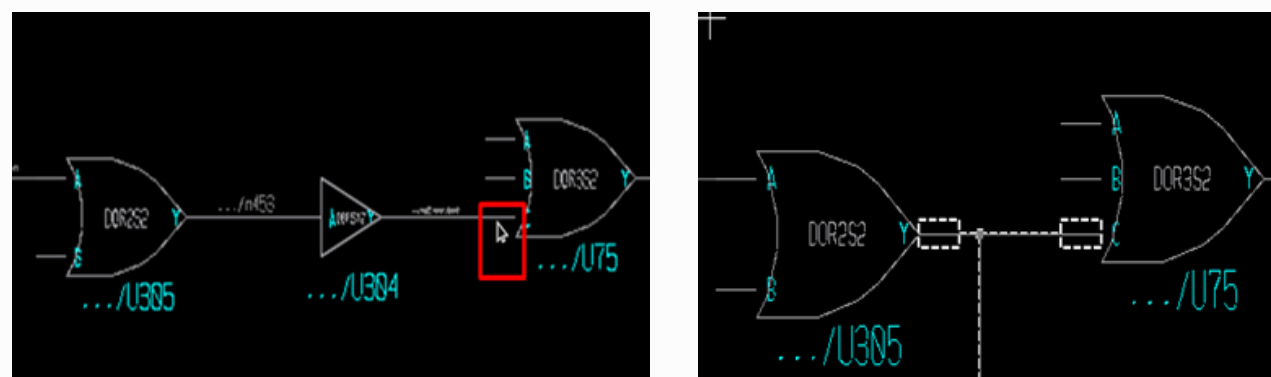
Design Related / How to Do Function ECO in Schematic View

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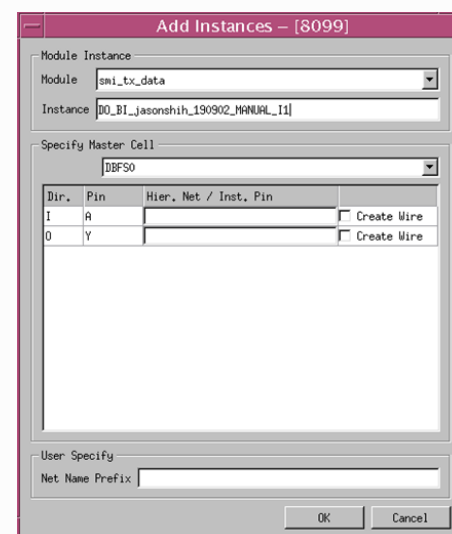
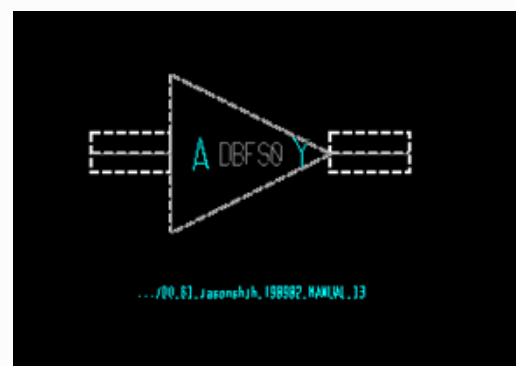
④ Change Driver :

Click  and select Sink pin, then click the new driver you want.



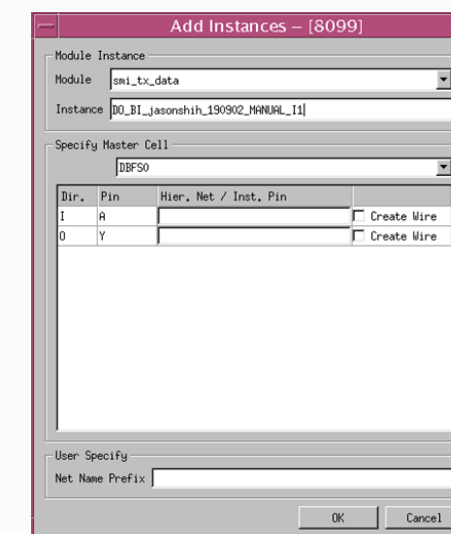
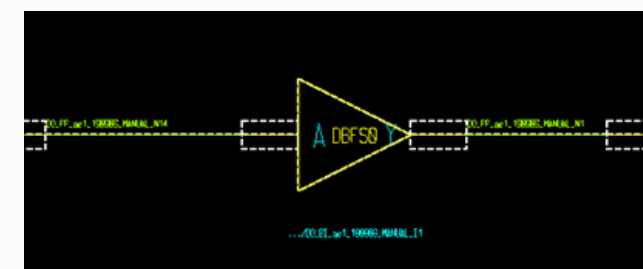
⑤ Create Instance :

Click  and specify module/instance name/master, and the new instance will be created.



⑥ Connect Instance pin :

Press  and click 2 instance pins which want to connect, and the new net will be created.



⑦ After function ECO, please go to “Netlist View” to write out ECO result.

Ref.

```
ecotclout -innovus ./ecout/eco_icc.tcl  
verilogout          ./ecout/
```

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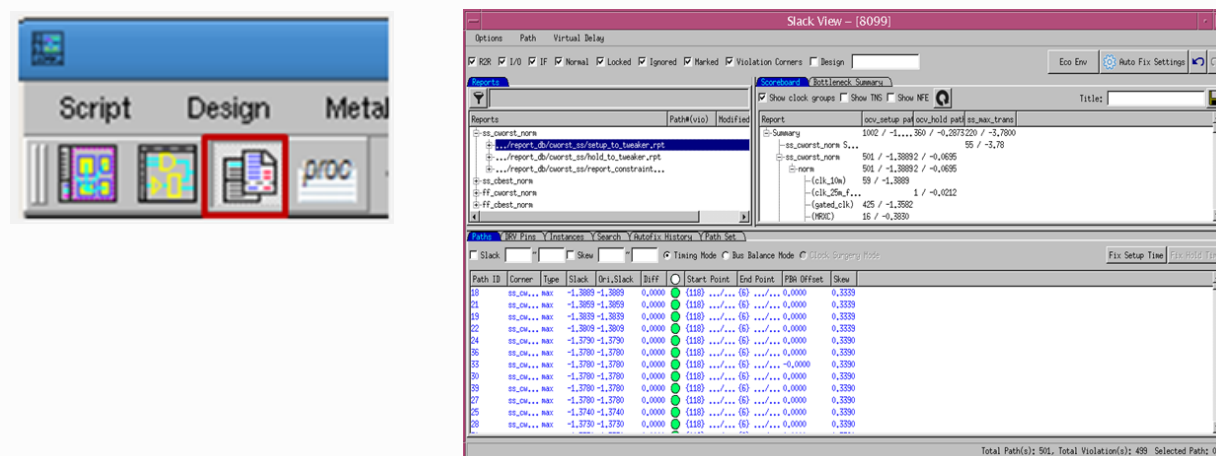
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Violation Related / How to Open Timing Violation Path

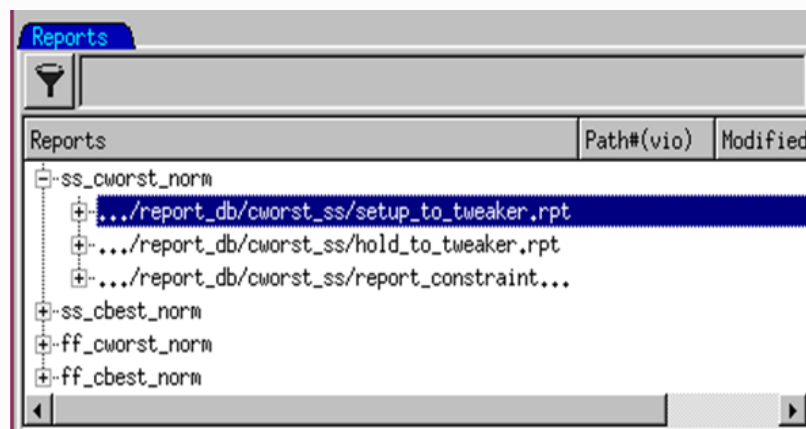
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- ① Use hot key “F8” or press “Slack View” button in Netlist view to open “Slack View” .



- ② In “Reports” , all types of report for each STA corner will be shown inside.



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Violation Related /

How to Open Timing Violation Path

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09.02.2019

③ “Scoreboard” will summarize total numbers of violations base on different type of ECO categories.

Report	ocv_setup pat	ocv_hold pat	ss_max_trans
Summary	1002 / -1....360	-0.2873220 / -3.7800	
ss_cworst_norm S...			55 / -3.78
ss_cworst_norm	501 / -1.38892	-0.0695	
norm	501 / -1.38892	-0.0695	
clk_10m	59 / -1.3889		
clk_25m_f...		1 / -0.0212	
gated_clk	425 / -1.3582		
(MRXC)	16 / -0.3830		

④ Double click on the path will open “Path View” for the specific path.

Path ID	Corner	Type	Slack	Ori.Slack	Diff	Start Point	End Point	PBA Offset	Skew
521	ss_cb...	max	-1.3889	-1.3889	0.0000	{118}	.../... {6}	0.0000	0.3339
18	ss_cw...	max	-1.3889	-1.3889	0.0000	{118}	.../... {6}	0.0000	0.3339
21	ss_cw...	max	-1.3859	-1.3859	0.0000	{118}	.../... {6}	0.0000	0.3339
524	ss_cb...	max	-1.3859	-1.3859	0.0000	{118}	.../... {6}	0.0000	0.3339
522	ss_cb...	max	-1.3839	-1.3839	0.0000	{118}	.../... {6}	0.0000	0.3339

Point	Master	Incr	Path
Startpoint: .../regs_p/apsel_reg/CK			
Endpoint: .../U_autoneg_txdata_bit/txword...			
Path Group: clk_10m			
Path Type: max			
clock G_write		90.0000	90.0000
clock network delay		1.0188	91.0188
dorado_data_path			
.../regs_p/apsel_reg/CK	DDFS1	0.0000	91.0188 r
.../regs_p/apsel_reg/Q	DDFS1	1.3620	92.3808 r
main0/LBUSTOPMod/regs_p/apsel	regs		
main0/LBUSTOPMod/apsel	ibustop		
main0/apsel	main		
U338/S0	IMUX2S2	0.0000	92.3808 r
U338/Y	IMUX2S2	0.7910	93.1718 f
U351/A	DMX2S2	0.0070	93.1788 f
U351/Y	DMX2S2	0.4060	93.5848 f
U350/A	DMX2S2	0.0000	93.5848 f
U350/Y	DMX2S2	0.6860	94.2708 f
U_dsub/rec_power_down	dsub		
U_dsub/U_phydig/purdn	phydig		
U_dsub/U_phydig/U_register/purdn	register		
.../U_register/U305/A	DMX2S2	0.0240	94.2948 f
.../U_register/U305/Y	DMX2S2	0.5110	94.8058 f
.../U_register/U304/A	DDFS12	0.0000	94.8058 f
clock clk_10m		100.0000	100.0000
clock network delay		0.6849	100.6849
clock uncertainty		-0.3000	100.3849
dorado_data_path		0.0000	100.3849
library setup time	null	-0.3470	100.0379
data required time			100.0379

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Violation Related /How to Review Blocking Code for Timing Violation

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09.02.2019



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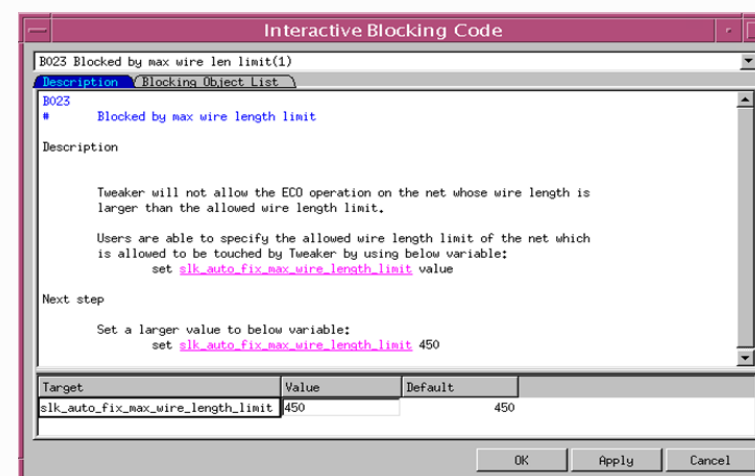


FAQ

- ① Review blocking code in “Path View”
Blocking code for each instance will shown in the column.
Double click the blocking code icon will show in detail.

[illegible]

- ② User can read blocking codes' description and relax/disable Tweaker setting to enhance ECO ratio.



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Violation Related /How to Review Blocking Code for Timing Violation

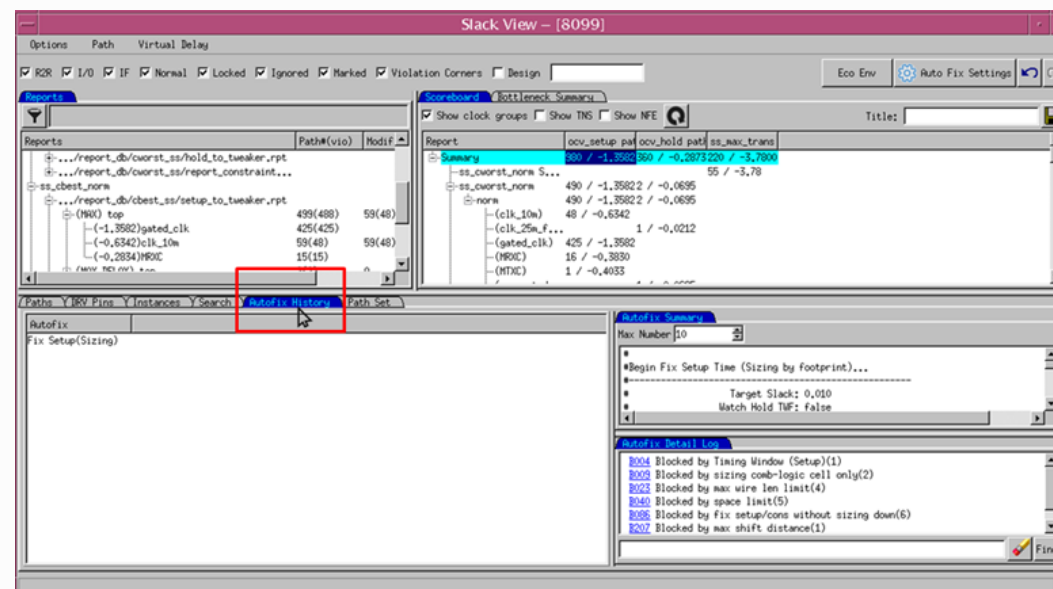
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③ Review blocking code in “Slack View”

Click “Autofix History” , every autofix result will shown in the window.

Double click the blocking code icon will show in detail.



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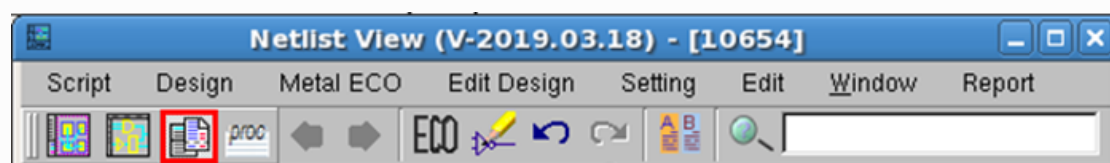
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Violation Related / How to Check Blocking Code in Detail

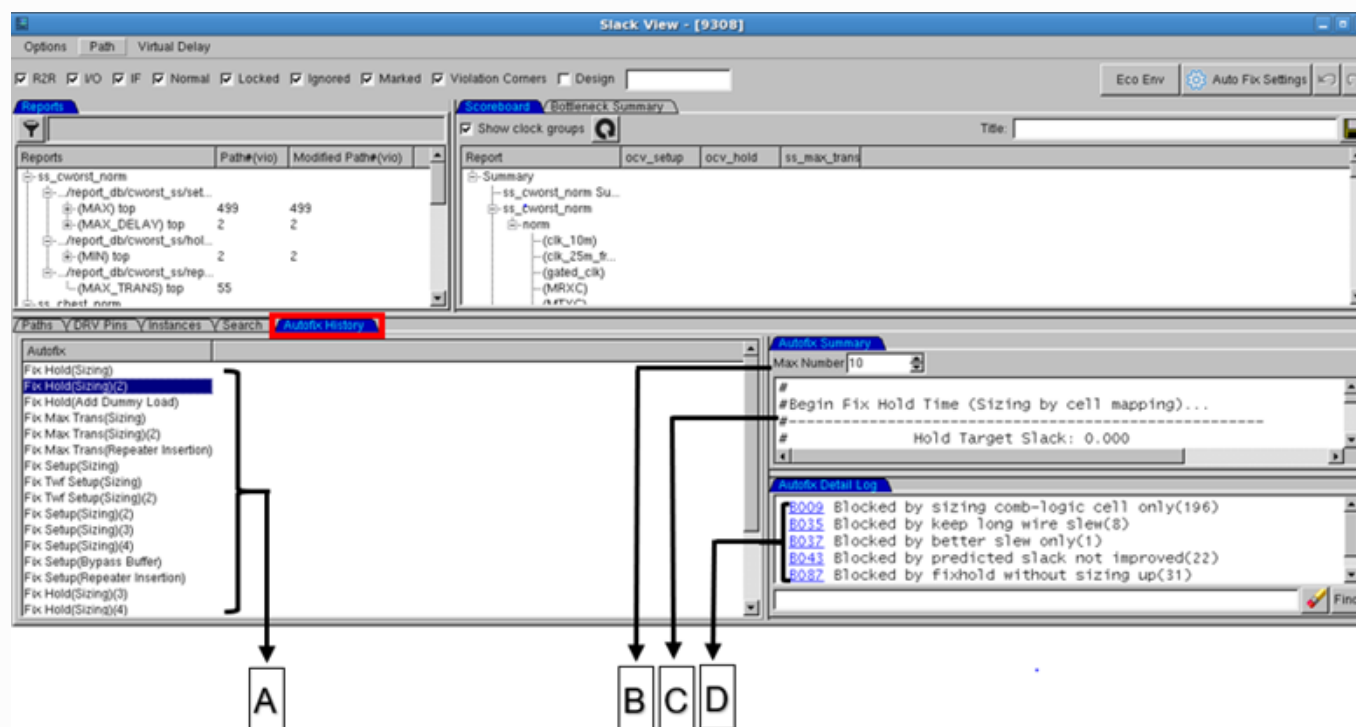
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09.05.2019

- ① Press "Slack View" icon in Netlist(main) View.



- ② Select the "Autofix History" tab in Slack View.



- A. List of fixing iterations which are executed in the run
- B. Max number of top moved cells
- C. When you click on any autofix in (A) It will help you to get the autofix summary of the specific iteration.
- D. In Parallel with (C) you will get the Blocking codes of (A)

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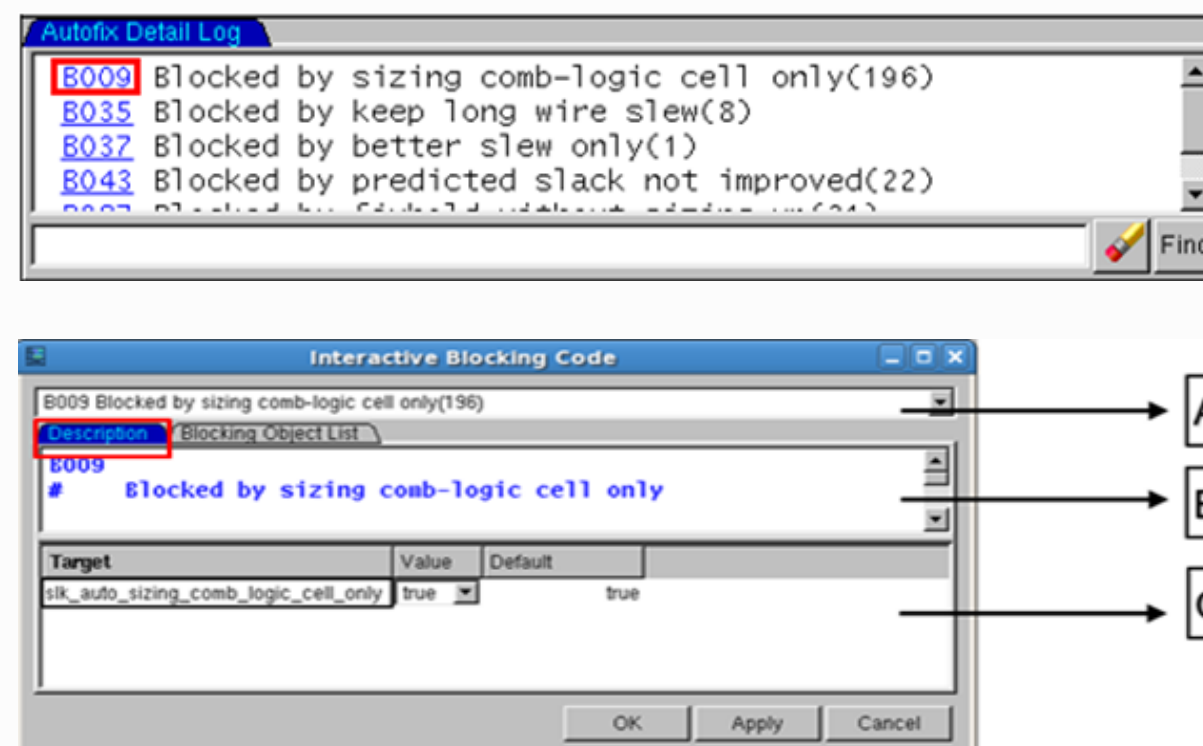
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Violation Related / How to Check Blocking Code in Detail

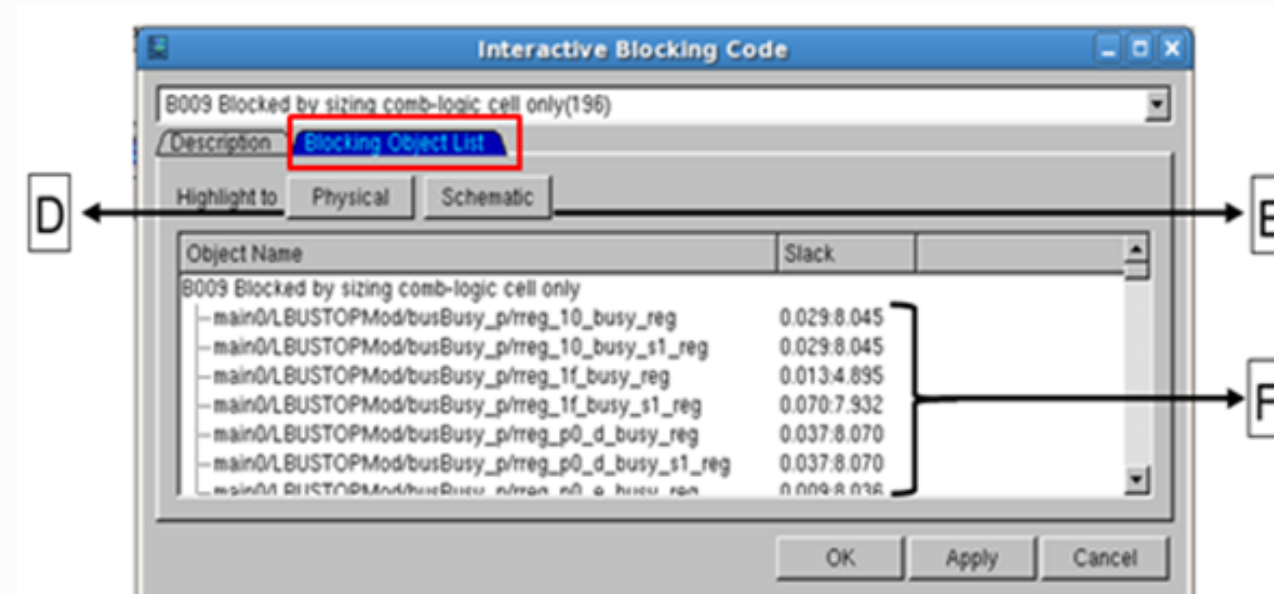
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09.05.2019

③ Checking Blocking Code Detail:

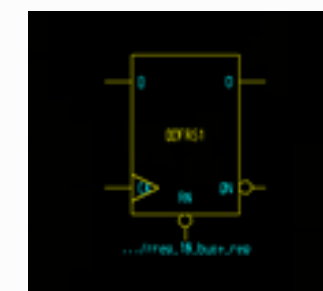


List of Blocked Objects:



- A. Blocking Code Name along with the number of affected nodes
- B. Description of the Blocking Code as well as it will suggest you to how you can relax that Blocking code
- C. List of Variables responsible for the Blocking code
- D. View the blocked object in the Physical View

- E. View the blocked object in the Schematic View
- F. Objects Blocked due to the Blocking Code with hold and setup slack.



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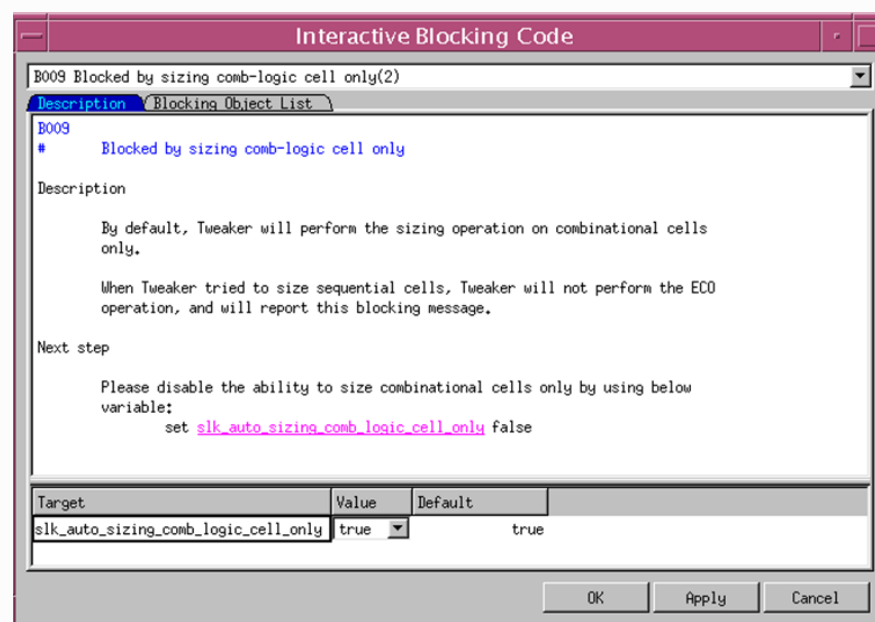
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Violation Related / How to Relax Blocking Code for Timing Fixing

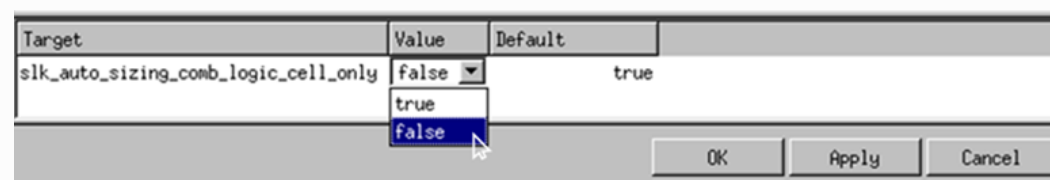
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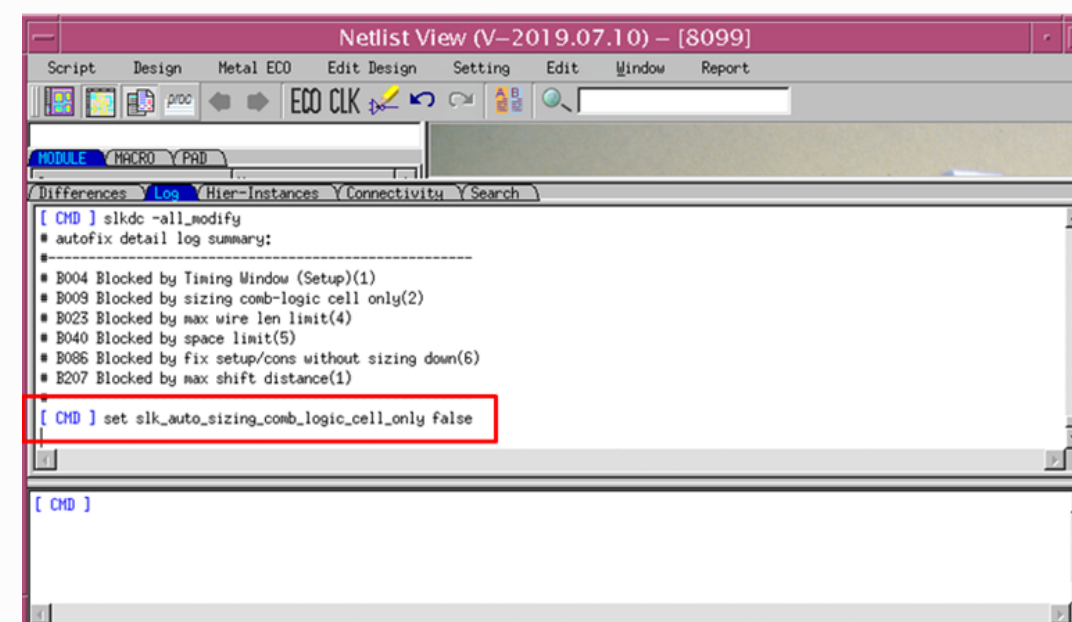
① Open “interactive blocking code” window.



② Change the suggested setting from the blocking code then click “OK” to apply the setting.



③ The setting command will shown in “Netlist View” .





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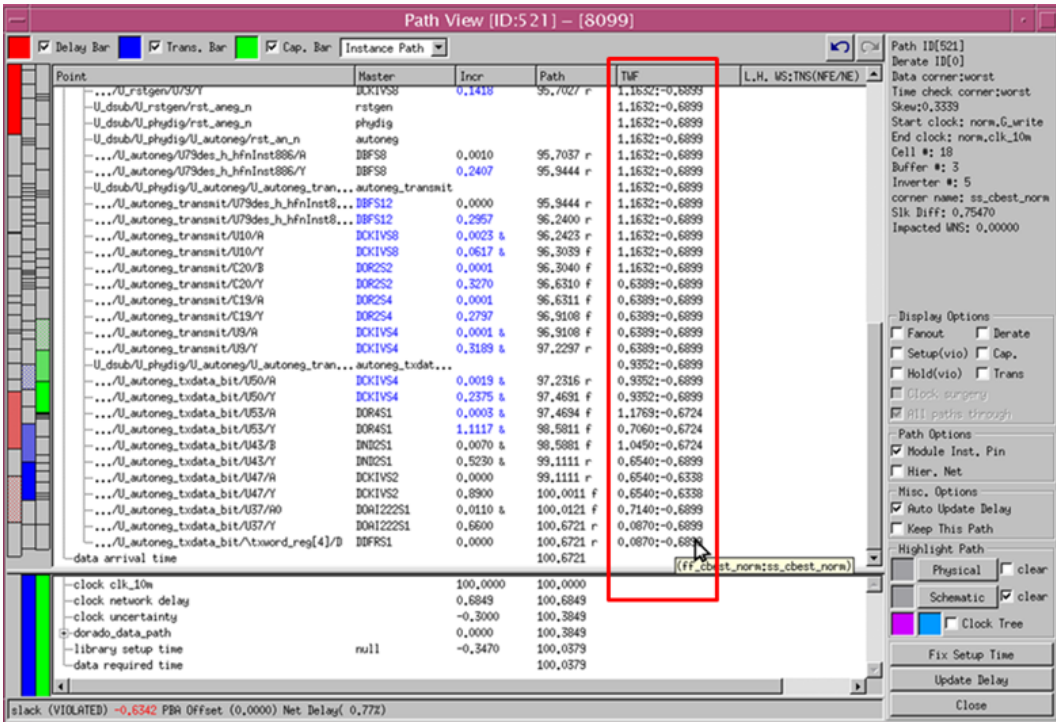
Timing Related /

How to Check Timing Margin

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09.03.2019

① Check timing margin in “Path View” (Hold margin : Setup margin).



② Move your mouse on the value will show the worst twf corner.



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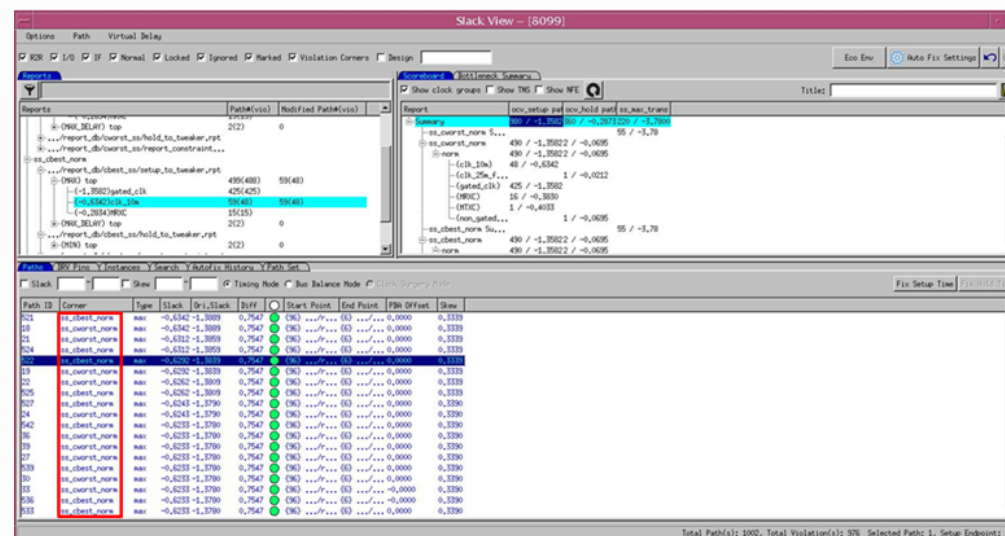
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Timing Related / How to Check the Worst Scenario

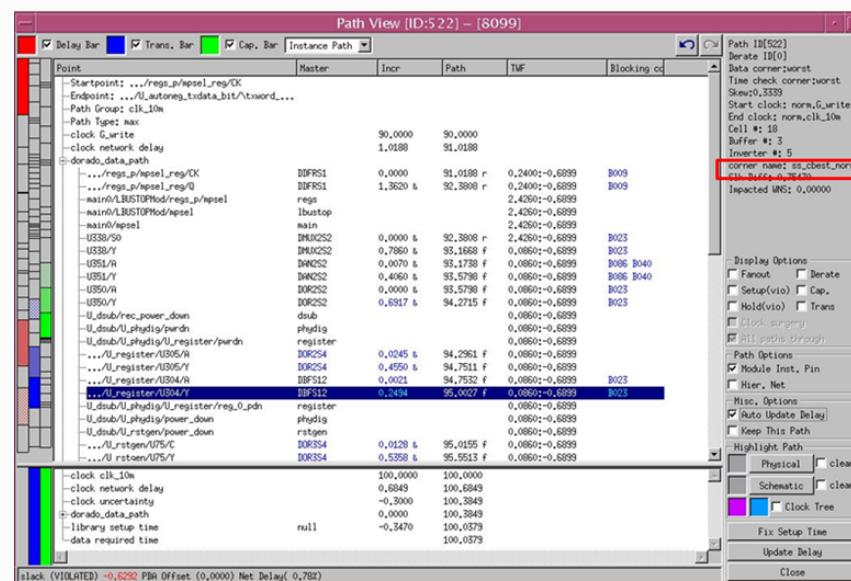
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09.03.2019

① Check scenario for violation path in “Slack View” .



② Check scenario for violation path in “Path View” .





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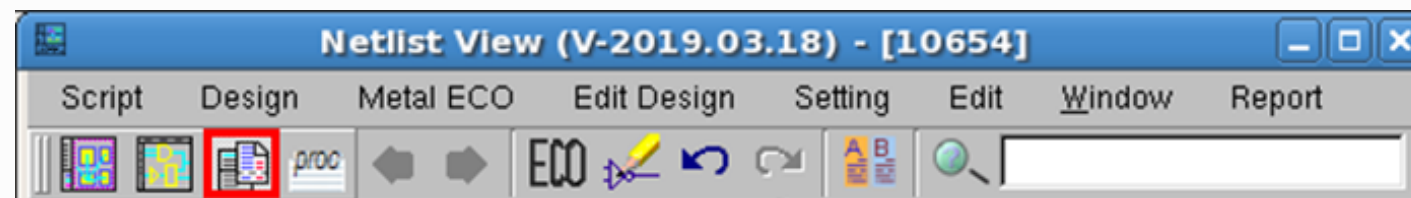
organized by Dorado

Timing Related / How to Check Timing Violation in Specified Slack Range

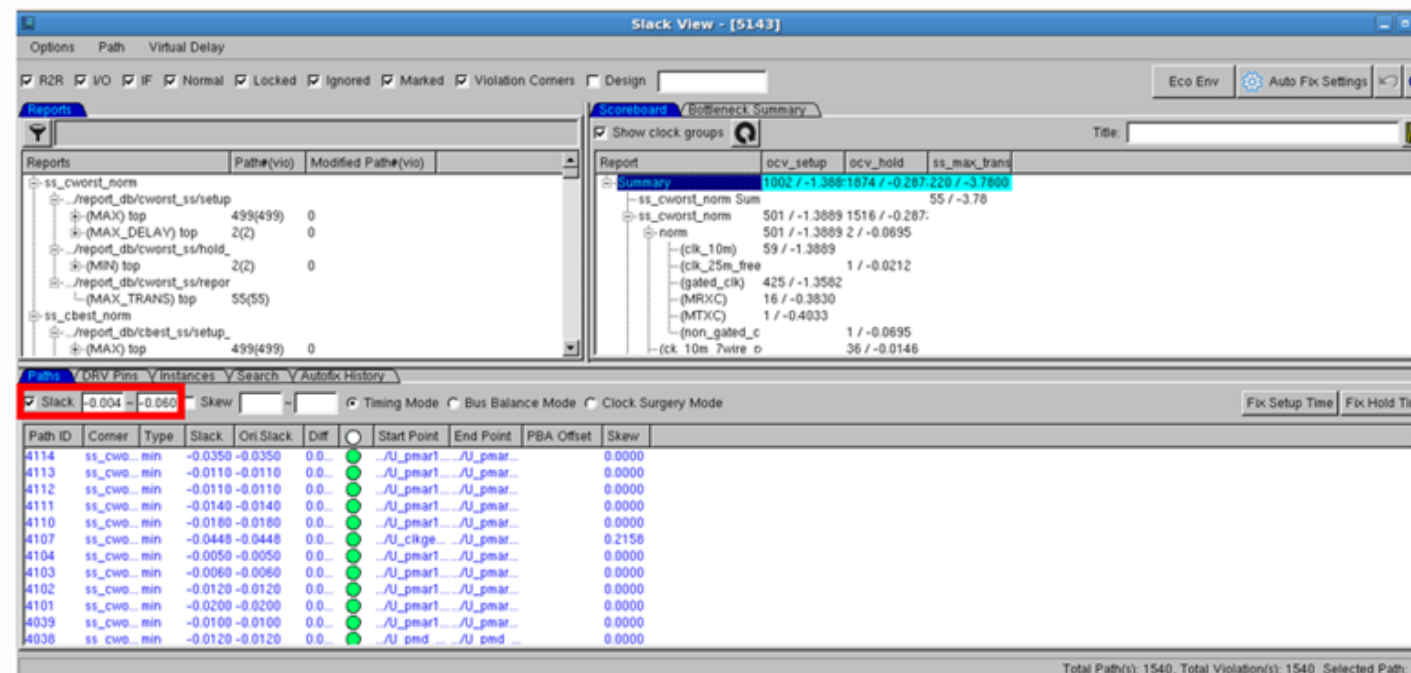
License Category: Tweaker-T1

09.03.2019

① Press “Slack View” icon in netlist(main) view.



② Check the ‘Slack’ check box and give the desired Slack Range to check the timing violations in a specific slack range in the Slack View.





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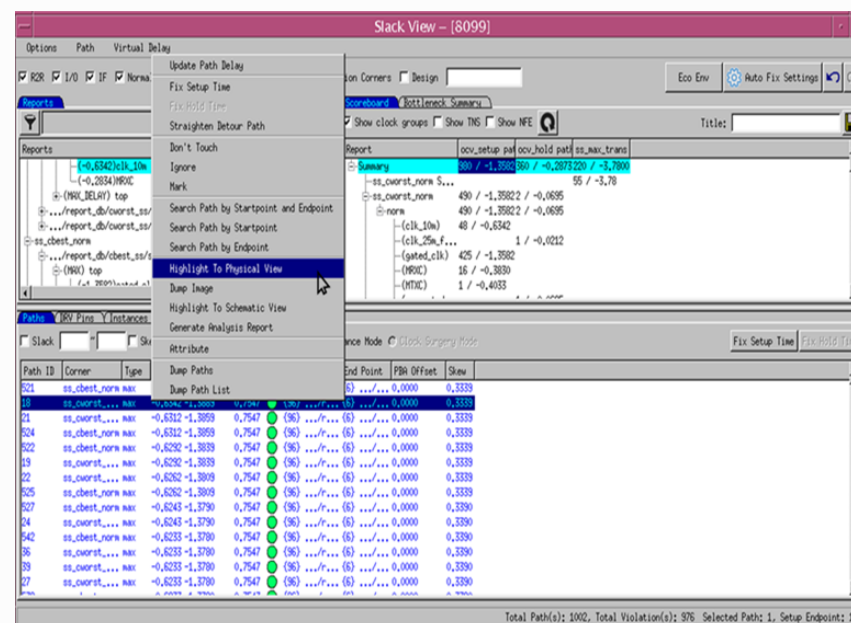
Physical Related / How to Highlight Timing Path in Physical View

License Category: Tweaker-T1

09.03.2019

① From “Slack View”

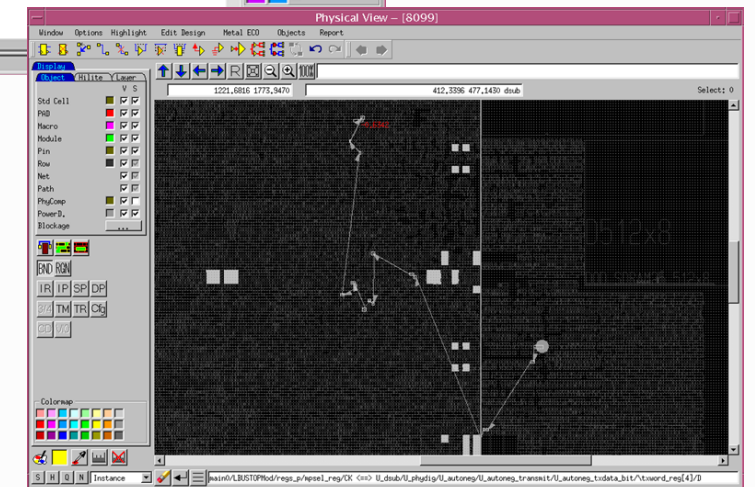
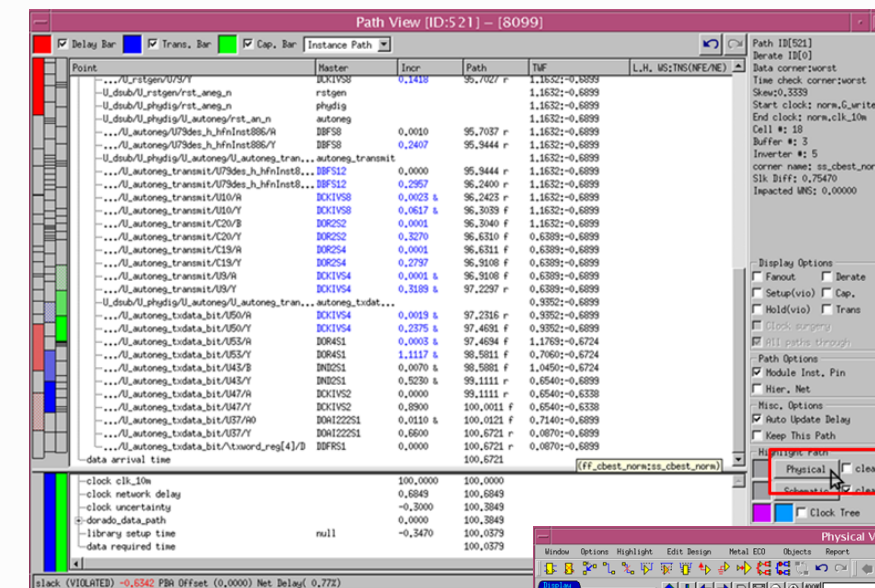
Right click on the path and click “Highlight To Physical View” .



② From “Path View”

Click “Physical” to show the path in “Physical View” .

(choose “clear” will remove previous highlight paths in “Physical View”)





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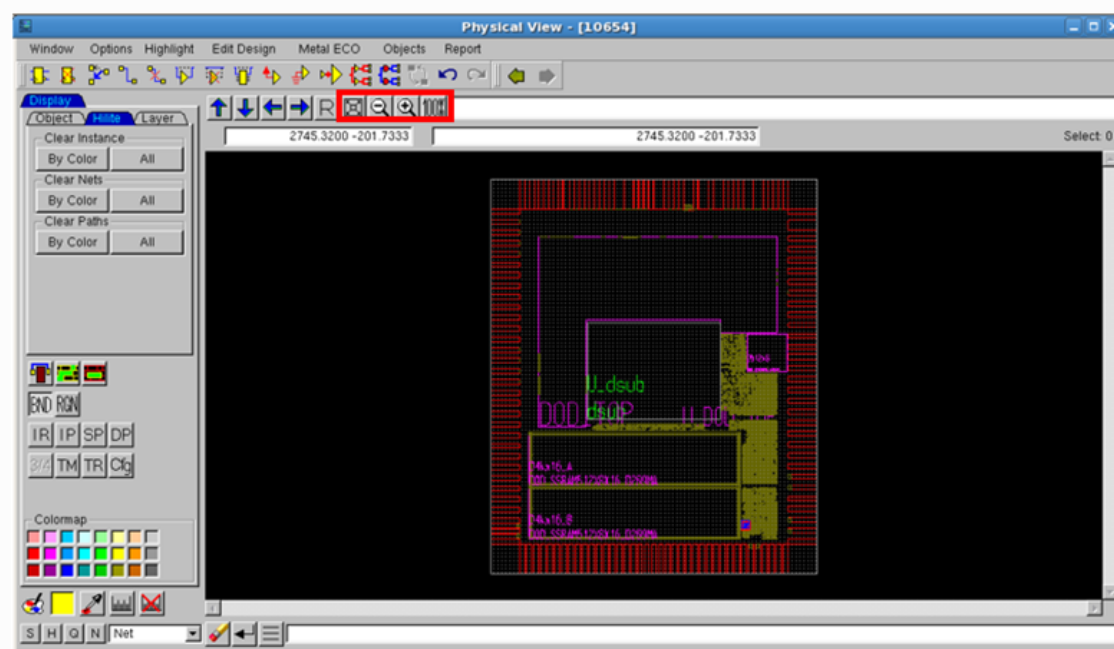
Physical Related / How to Zoom-in and Zoom-out in Physical View

License Category: Tweaker-T1

09.05.2019

Method I:

Zoom In/Out Buttons in Physical View.



Zoom In(Hot Key : Z):

Click on  button to magnify the view of a design in the Physical View.

Zoom Out (Hot Key : z):

Click on  button to reduce the view of a design in the Physical View.

Fit to Screen (Hot Key : F):

Click on  button to display the entire design as large as possible (fit to the screen) in the Physical View screen.

Method I:

Go to "Options" -> "Settings" from the menu bar of Physical View.



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Physical Related / How to Zoom-in and Zoom-out in Physical View

License Category: Tweaker-T1

09.05.2019

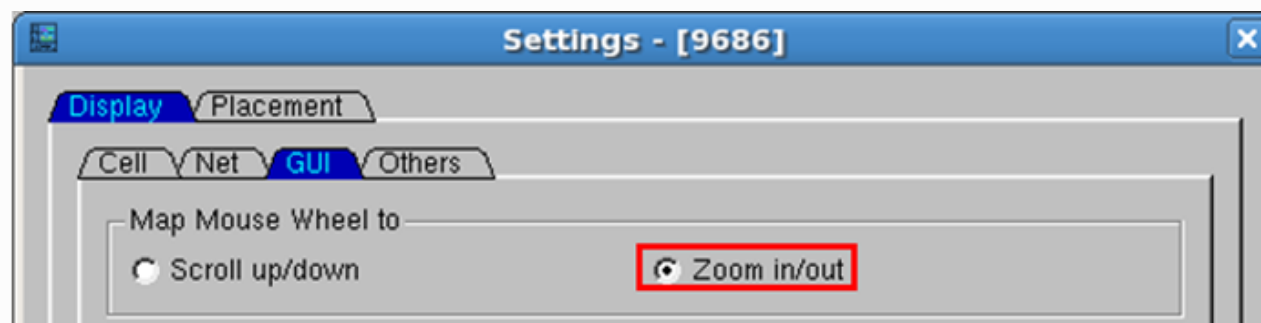
Method I:

Go to "Options" -> "Settings" from the menu bar of Physical View.



In the settings window, under "GUI" tab, select Map Mouse Wheel to "Zoom in/out"

Scroll the mouse wheel to perform zoom in/out in the Physical View.



Method III:

In Physical View, right button of the mouse can also be used to zoom in/out.

This method is same as 'Zoom in by Window' from Method I but here right mouse button will be used instead of left.

Zoom In: Right click on any point of the design and drag the cursor **from left to right** while pressing the right mouse button, release the button after selecting the area you want to zoom in.

Zoom Out: Right click on any point of the design and drag the cursor **from right to left** while pressing the right mouse button, release the button to zoom out.



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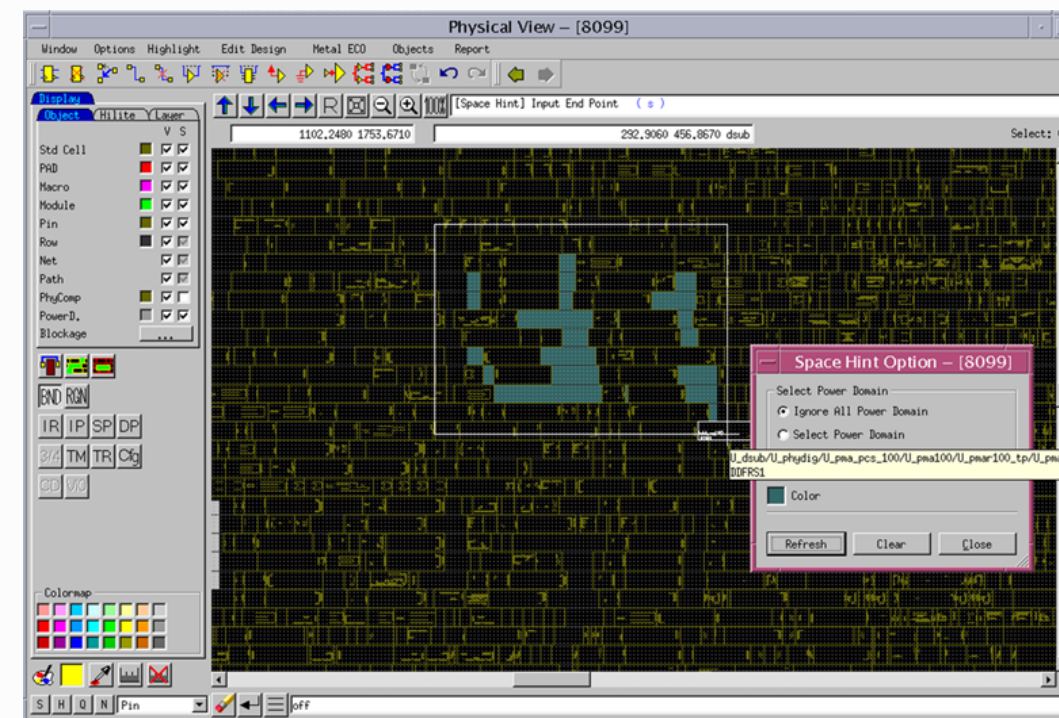
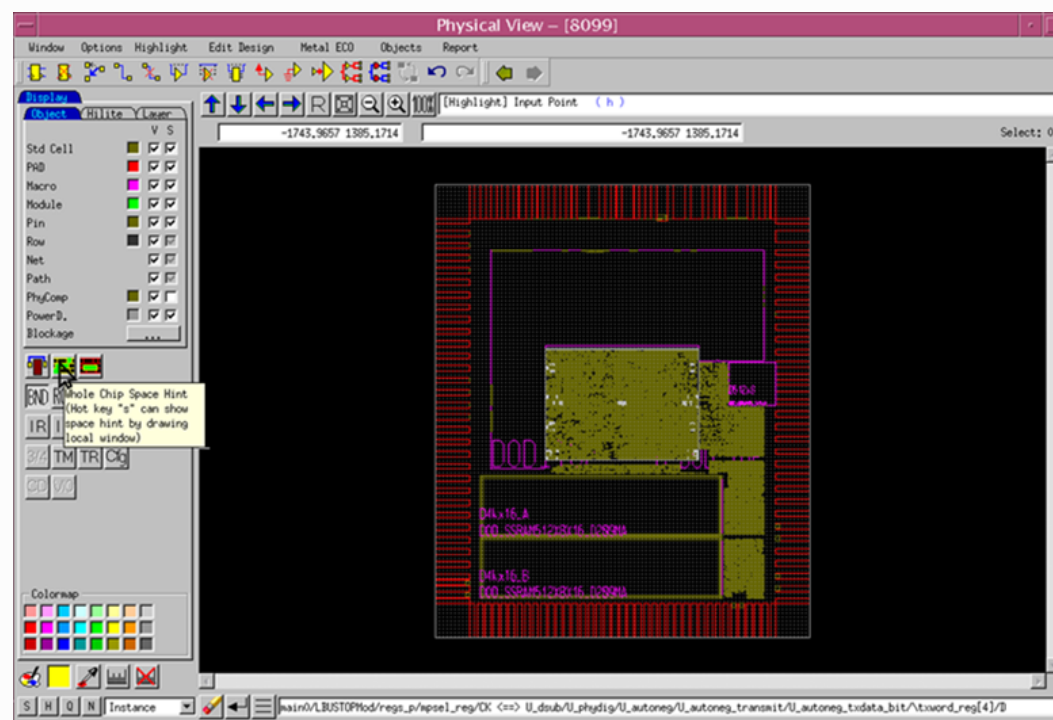
organized by Dorado

Physical Related / How to Highlight Available Resource

License Category: Tweaker-T1

09.03.2019

- ① Left click  to highlight available resource in “Physical View” . ② Use hot key “s” and select the window to highlight free space in “Physical View” .





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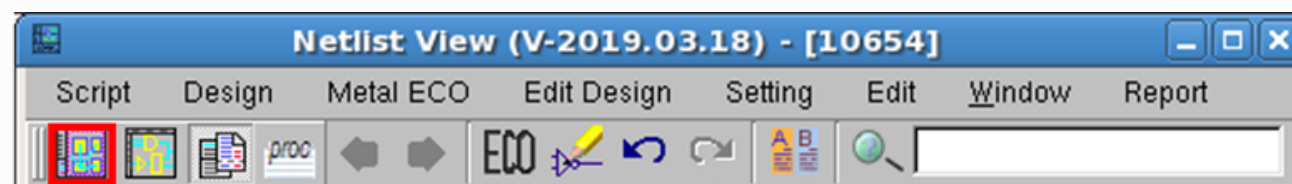
organized by Dorado

Physical Related / How to Highlight Detail Route in Physical View

License Category: Tweaker-T1

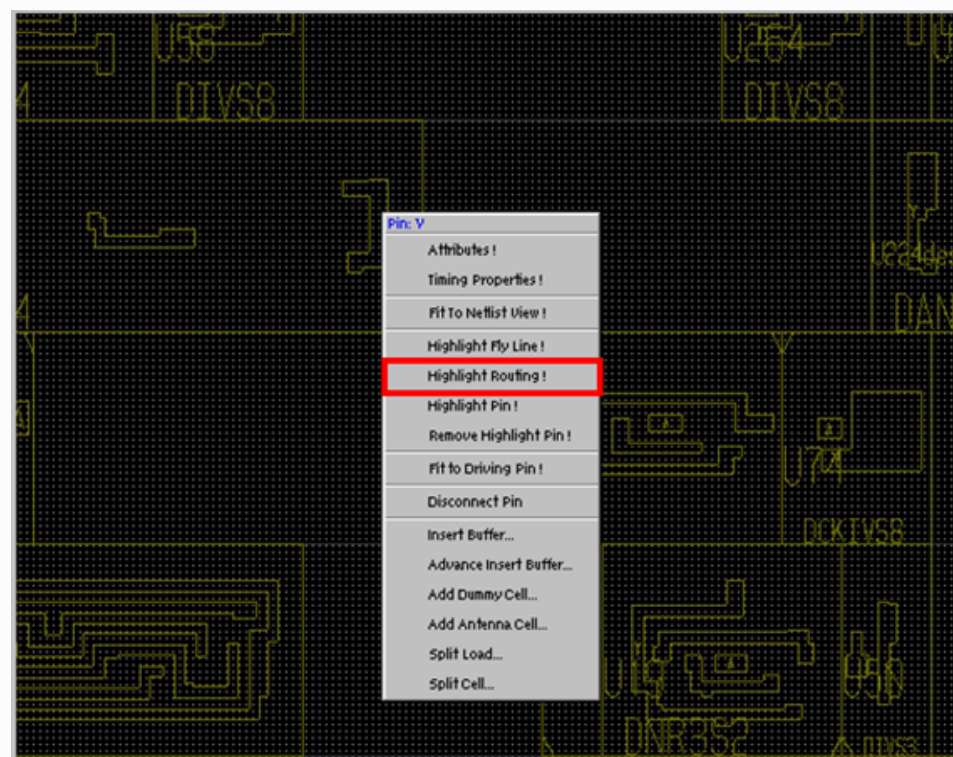
09.05.2019

Press “Physical View” icon in netlist(main) view.

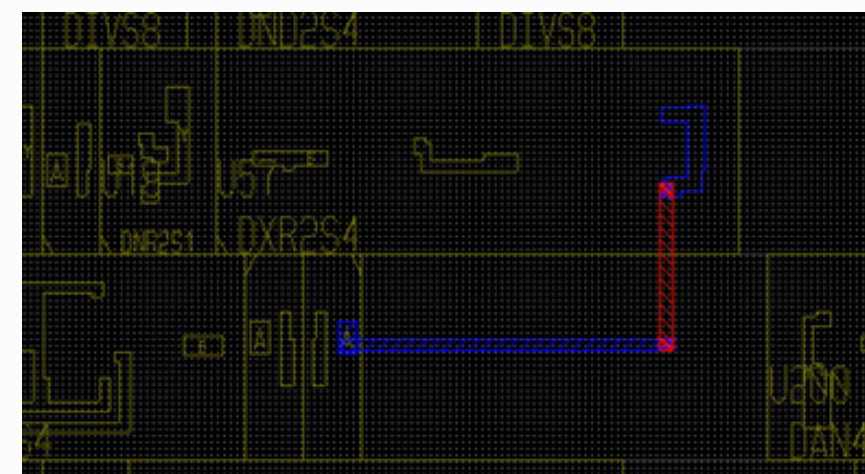


Highlighting Detail Route:

① Right click on any pin in the design from Physical View and click on “Highlight Routing!” .



② Highlighted Routing from the selected pin.



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Physical Related / How to Highlight Detail Route in Physical View

License Category: Tweaker-T1

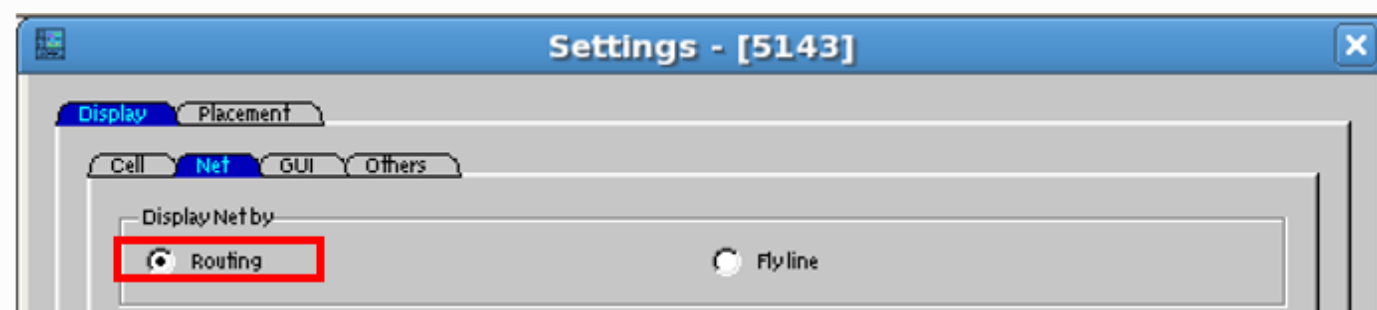
09.05.2019

Alternative Method for Highlighting Detail Route:

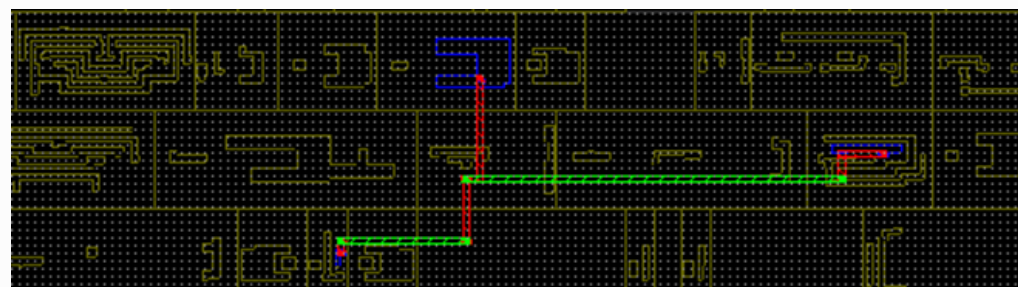
① Go to 'Options' -> 'Settings' from the menu bar of Physical View.



② In the settings window, under "Net" tab, select Display Net By "Routing".



③ Double click on any pin in the design from the Physical View to Highlight Routing from that pin.



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Physical Related / How to Highlight Detour Net Physical View

License Category: Tweaker-T1

09.03.2019

report_detour_net # Tweaker command to report the detour net by desired options.

report_detour_net command usage :

report_detour_net [options]

[options]:

-file file_name

-length min_length # default 40 um

-length_ratio ratio # default 1.3(130%)

-cap_ratio ratio # default 1.3(130%)

-input_net input_file

-highlight

-verbose

-domain path_setup | twf_setup | path_hold | max_trans | max_cap | si | power

-sort setup_slack | length_ratio # default detour_ratio

-fanout_less_than fanout count # default 3

-slack_range thresholdL thresholdH

-include_clock_net

-corner corner_name_list

-skip_corner corner_name_list

-min_net_delay delay # default 0.02

SEE
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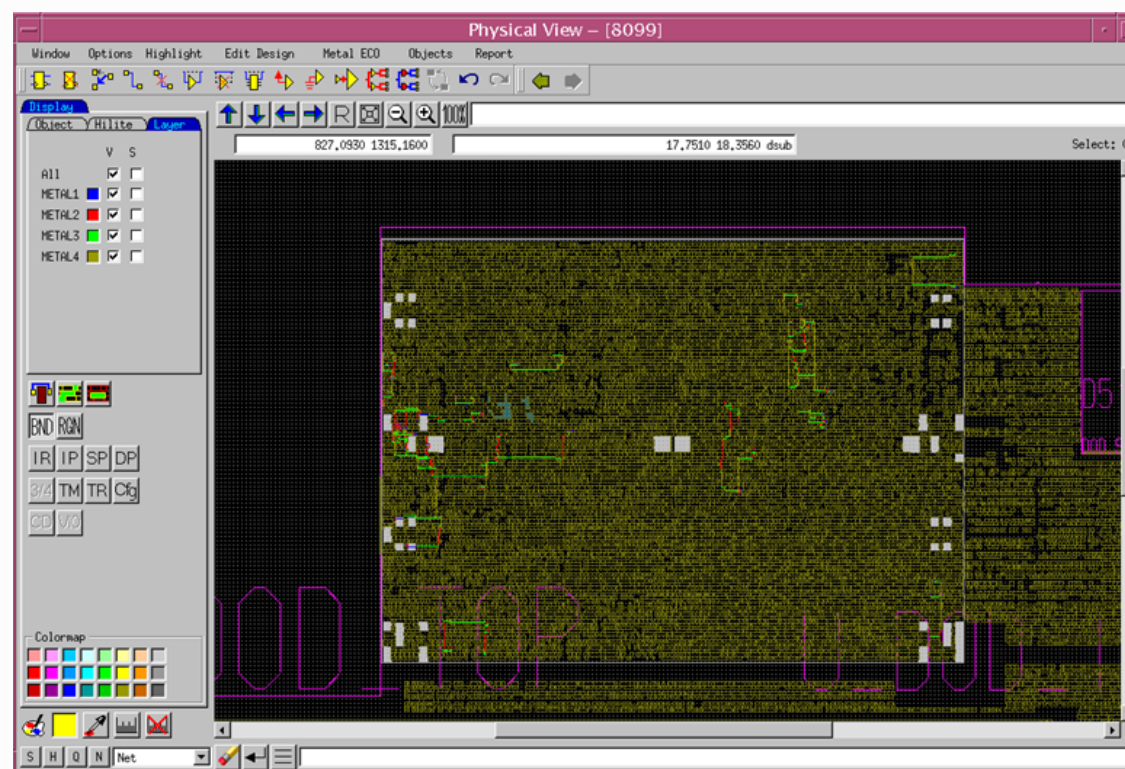
organized by Dorado

Physical Related / How to Highlight Detour Net Physical View

License Category: Tweaker-T1

09.03.2019

Ex: report_detour_net -highlight -length_ratio 1.3 -length 40 -cap_ratio 1.3



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Physical Related / How to Highlight Physical Rule Violation

License Category: Tweaker-T1

08.31.2019

In FinFET design, Tweaker will consider FinFET physical rule during ECO stage. Tweaker also provides the command for the user to check if there is any instance violate in the physical rule.

1. To report physical violation

```
tweaker_shell> help report*violation
```

2. To check physical violation in Tweaker physical view

a. Enable internal variable and use Physical View to debug physical violation

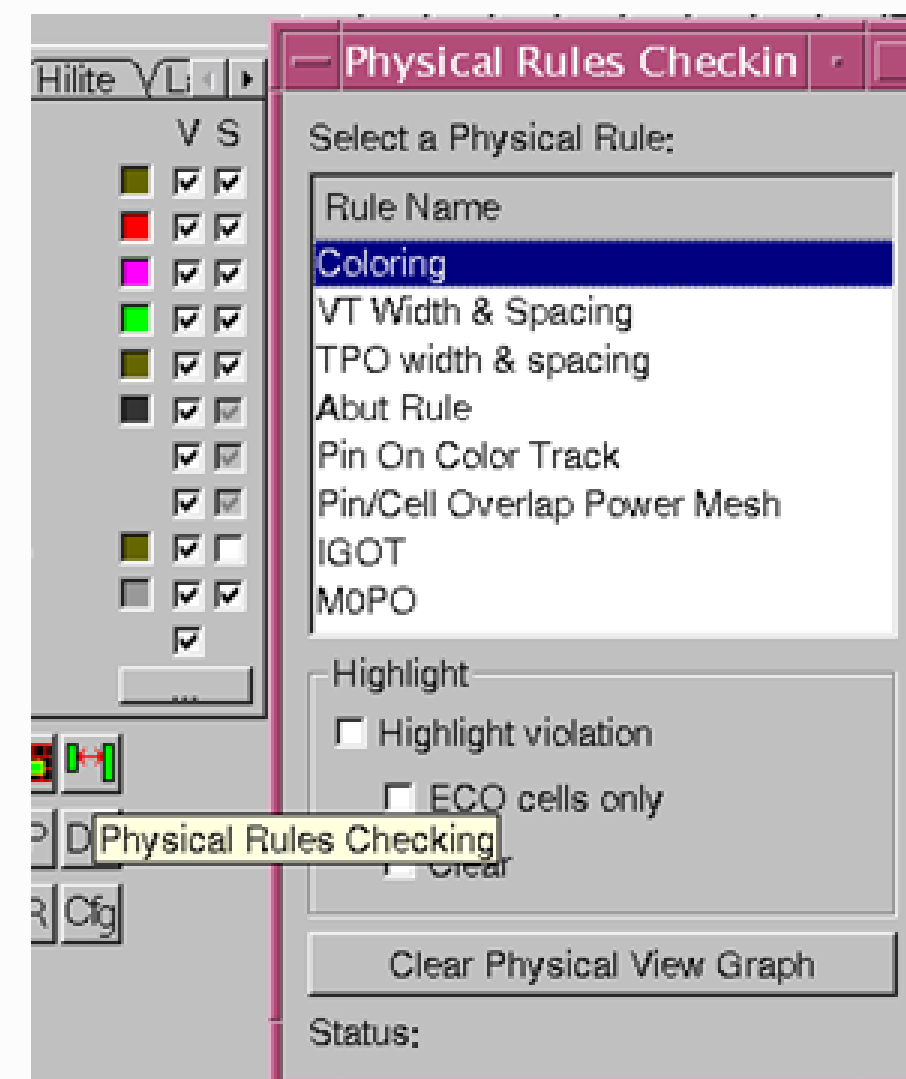
```
set_enable_physical_rules_checking_gui true
```

b. A new icon will be shown on Physical View

Step 1: Click the physical rule

Step 2: click the instance

Step 3: check if the rule is applied





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FAQ

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Log Message / How to Check ECO Environment Setting

License Category: Tweaker-T1

08.31.2019

Descript:

in Tweaker log file, user can sear key word “Begin” to review Tweaker environment setting before ECO start

Ref.

#Begin Fix Hold Time (VT Swapping)...

```
#-----  
#      Hold Target Slack: 0.000  
#      Setup Target Slack: 0.000  
#      Watch Setup TWF: true  
#      Watch Hold TWF: true  
#      Watch Clock As Data: true  
#      Min Improved Slack: 0.002  
#      Auto Sizing High Effort: false  
#      Fit To Free Space: true  
# Max Shift Distance For Sizing: 0.000 micron  
# Auto Sizing Max Fanout Limit: 999  
# Auto Fix Max Wire Length Limit: 9999 micron  
#      Enable SI Aware: false  
#      Fix SI Only: false  
#      Fix SI First: true  
#      Fix Hold By Fix Setup SI: false  
# Fix Hold By Extract Setup Margin: false  
#      Fix Hold Without Sizing Up: true  
#-----
```

#autofix max loop(10)

#fix design(\$eco_blocks)...



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Log Message / How to Check ECO QoR in Each Iteration

License Category: Tweaker-T1

08.31.2019

Descript:
in Tweaker log file, user can sear key word “End” to review Tweaker QoR base on each of ECO iteration

Ref.

#

#End Fix Hold Time (VT Swapping)

#-----

File: hold_to_tweaker.rpt.gz

group(s): PCLK,async_default,tck

#-----

sizing count: 27

autofix summary:

setup vio paths: (58720 -> 58720)

setup TNS(vio paths): (-22020.918 -> -22020.844)

setup worst slack: (-117.331 -> -117.331)

setup endpoints: (28030 -> 28030)

setup TNS(endpoints): (-6244.035 -> -6244.030)

#

hold vio paths: (7675 -> 7663)

hold TNS(vio paths): (-116.017 -> -115.380)

hold worst slack: (-0.940 -> -0.940)

hold endpoints: (7675 -> 7663)

hold TNS(endpoints): (-116.017 -> -115.380)

#

#-----

path status:

total setup vio paths: (58720 -> 58720)

total setup TNS(vio paths): (-22020.918 -> -22020.844)

total setup worst slack: (-117.331 -> -117.331)

total setup endpoints: (28030 -> 28030)

#

total hold vio paths: (7675 -> 7663)

total hold TNS(vio paths): (-116.017 -> -115.380)

total hold worst slack: (-0.940 -> -0.940)

total hold endpoints: (7675 -> 7663)

run time: 00:03:19

#-----

[CMD] slkdc -all_modify

autofix detail log summary:

#-----

B004 Blocked by Timing Window (Setup)(27)

B005 Blocked by Timing Window (Hold)(20)

B026 Blocked by Timing Window (Hold) (Input Pin)(13)

B034 Blocked by DRV fail(58)

B043 Blocked by predicted slack not improved(114)

B051 Blocked by don't use setting(15)

B087 Blocked by fixhold without sizing up(166)

B173 Blocked by incorrect DEF file, please use "check_def_quality"(10568)

#-----



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Log Message / How to Check Detail about Tweaker Blocking Code or Error Code

License Category: Tweaker-T1

09.06.2019

For all of Tweaker Blocking code or Error code, it will show the example as below

Ex: B001 or E00010

Users can type “**man B001**” in command window to check in detail and the next step.



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Script Information / How to Check Detail in Tweaker Command and Variable

License Category: Tweaker-T1

09.06.2019

For all of Tweaker variable, please use “**printvar**” or “**man**” to check variable current/default setting and detail information.

Ex:

```
printvar slk_drv_factor
```

For all of Tweaker command, please use “**help**” or “**man**” to check command syntax, description, or example.

Ex:

```
help set_drv_value
```



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Script Information / How to Read Timing Violations Which Only Related to BlockB

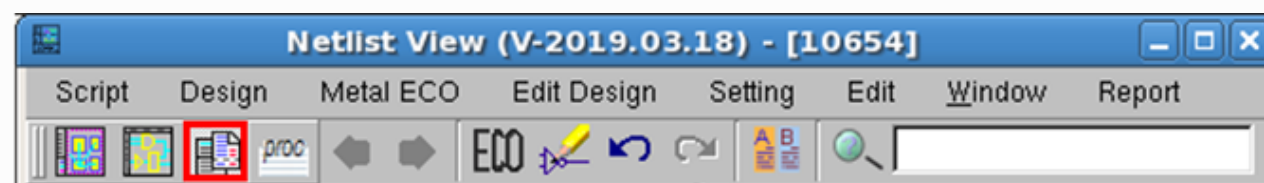
License Category: Tweaker-T1

09.05.2019

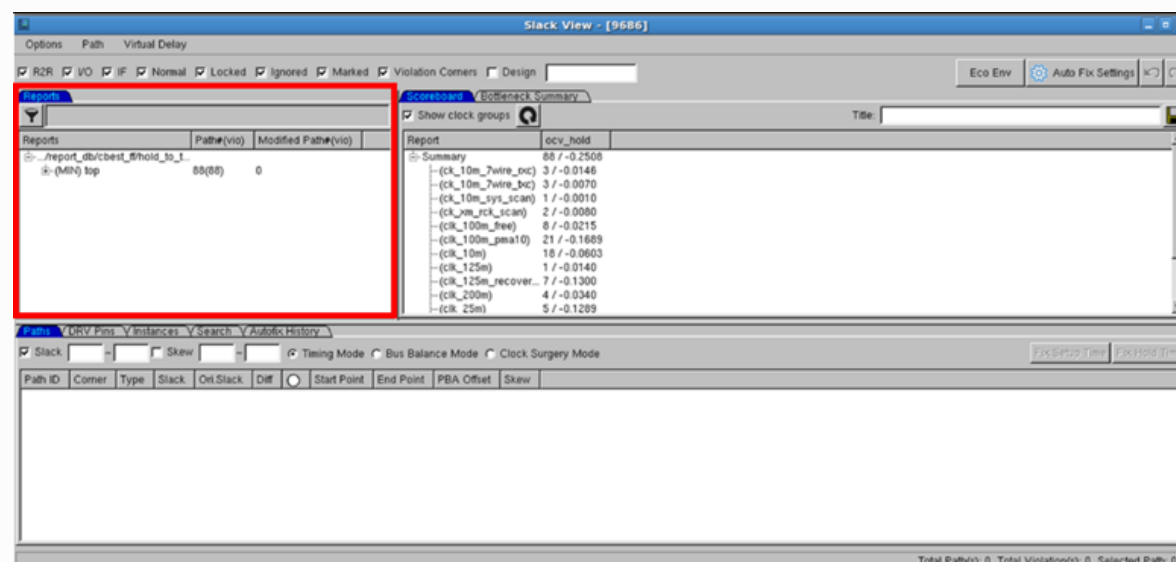
To Read Timing Information of a block we need to use 'slackin' command.

Usages: slackin <slack_file> -design_list <design name whose timing information are to be read>

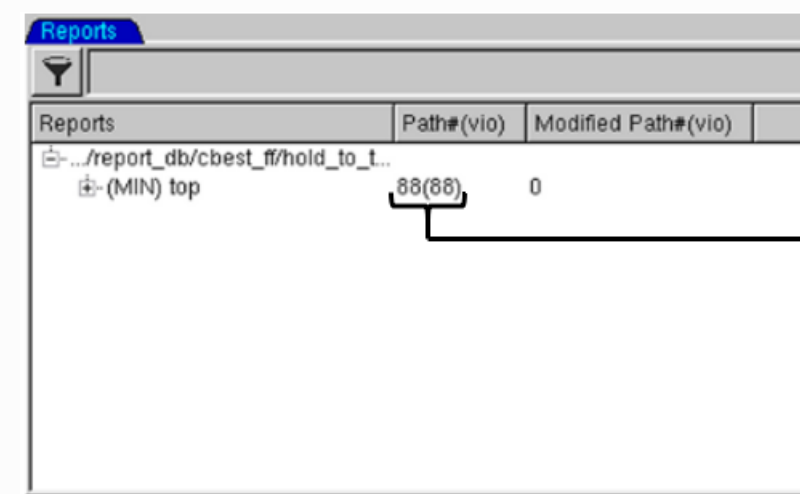
- ① Press "Slack View" icon in netlist(main) view.



- ② Slack View:



- ③ Check the Report window inside Slack View for violated path information.



Total Number of Paths in the given design/block and the number of violated paths.



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Script Information / How to Apply Don't Touch Setting

License Category: Tweaker-T1

08.31.2019

Descript:

Tweaker provides don't_touch and don't_use command base on instance, cell, net, pin.

1. To apply dont_touch base on instance
set_dont_touch_instance true TOP/sub_1/inst_a
2. To apply dont_touch base on cell
set_dont_use_cell true DBFS0LVT
3. To apply dont_touch base on pin
set_dont_touch_pin true TOP/sub_1/inst_buf_1/Y
4. To apply dont_touch base on nets
set_dont_touch_net true TOP/sub_1/net_002
5. To apply dont_user base on cells
set_dont_use_cell true DBFS0LVT

Related:

more related commands, you can check

- a. twk> help set_dont_touch*
twk> help set_dont_use*

more command usage, you can check

- a. twk> man set_dont_touch_instance



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Script Information / How to Fix Interface Timing Violation

License Category: Tweaker-T1

08.31.2019

Descript:

Tweaker provides option to fixing interface violation which timing paths cross the different hierarchical design.

1. To fix interface setup Reg2Reg violation

```
twk> slkfix -setup -interface -ff
```

2. To fix BlockB' s related interface violation

```
twk> slkfix -setup -interface -design_list { $blockB }
```

3. To fix interface violation from blockA to blockB only

```
twk> slkfix -setup -from_design{ $blockA } -to_design{ $blockB }
```

Related:

more command usage, you can check

a. twk> man slkfix